

Two-scale Beckner inequalities via power-divergence score transport

Candidate solution packet for arXiv:2509.15410

11 August 2026

Status: candidate full solution, likely valid, awaiting human review. For every $1 < p \leq 2$, the packet proves joint and mixture Beckner inequalities under a natural score-transport condition. Pointwise bounded scores give exactly the constants anticipated in the source; the source MGF condition gives the same formulas after replacing $\bar{L}^2$ by $p\bar{L}^2$.

1 Source question

The source is Vishwak Srinivasan, *Understanding two-scale criteria for Poincaré and log-Sobolev inequalities in the Euclidean case through Φ -entropies*, arXiv:2509.15410 [1]. In the Discussion and Section 6.1 (source PDF pages 21–22), the author asks whether the two-scale analysis extends to

$$\Phi_p(t) = \frac{t^p - 1}{p - 1}, \quad 1 < p \leq 2,$$

whose Φ -Sobolev inequality is the Beckner inequality. The source surmises that the joint law ν and mixture μ should inherit Beckner inequalities, but explains that its Legendre-duality proof leaves two nonzero terms and that Φ_p is not Legendre at zero.

Figures 1 and 2 reproduce the complete two-page passage.

Transcription of the mathematical question. Suppose the mixing law ρ satisfies $\Phi_p \text{SI}(\alpha)$ and every conditional law P_y satisfies $\Phi_p \text{SI}(\beta)$. Under an appropriate two-scale score criterion, do the joint law $\nu(dx, dy) = P_y(dx)\rho(dy)$ and the mixture $\mu(dx) = \int P_y(dx)\rho(dy)$ satisfy Beckner inequalities with constants analogous to the Poincaré and log-Sobolev constants in the source?

2 Setup and exact score criterion

We use the smooth Euclidean setup of the source. Thus ρ is a probability measure on $\mathbb{R}^{d_y}$ and P_y has a positive density p_y on $\mathbb{R}^{d_x}$. Differentiation under the integral is assumed whenever it is used, as in the source. Define the conditional score

$$s_y(x) = \nabla_y \log p_y(x), \quad \mathbb{E}_{P_y} s_y = 0.$$

For a positive function f , write

$$\text{Ent}_{\Phi_p, P}(f) = \mathbb{E}_P \Phi_p(f) - \Phi_p(\mathbb{E}_P f) = \frac{\mathbb{E}_P f^p - (\mathbb{E}_P f)^p}{p - 1}.$$

We say that P satisfies $\Phi_p \text{SI}(\gamma)$ when

$$\text{Ent}_{\Phi_p, P}(f) \leq \frac{\gamma}{2} \mathbb{E}_P [\Phi_p''(f) \|\nabla f\|^2], \quad \Phi_p''(t) = pt^{p-2}. \quad (1)$$

6 Discussion

In this note, we showed Poincaré and log-Sobolev inequalities for joint and mixture distributions induced by a marginal distribution and a collection of conditional distributions under certain multiscale criteria. In particular, we achieved this by viewing these functional isoperimetric inequalities as special case as Φ -Sobolev inequalities. We find that the multiscale criteria for the Poincaré and log-Sobolev inequalities ([Var](#) and [MGF](#), respectively) are fundamentally connected to the variational forms of their corresponding Φ -entropies. Through this more general analysis, we identify a previously unknown result about the mixture μ when $x \mapsto \nabla \log p_{X|Y=y}$ is Lipschitz-continuous. Some open questions that remain are as follows. Does this analysis more general apply to obtain Φ -Sobolev inequality that is neither the Poincaré nor the log-Sobolev inequality. We discuss the case of $\Phi_p(t) : t \mapsto \frac{t^p-1}{p-1}$ below where we face certain difficulties. The discussion about exact HMC motivates studying the other regularity properties of the Hamilton flow, which would enable a more complete picture about how isoperimetric properties evolve along exact HMC for strongly log-concave and smooth target distributions more generally.

6.1 Comments about interpolating between ent and var

Motivated by the generalising beyond the Poincaré and log-Sobolev inequalities, we investigate the case where $\Phi_p(t) = \frac{t^p-1}{p-1}$ for $p \in (1, 2]$. The associated Φ -entropy $\mathbb{J}^{\Phi_p}$ interpolates between [ent](#) and [var](#); the former is the limit of $\mathbb{J}^{\Phi_p}$ as $p \rightarrow 1$, and the latter is [var](#) when $p = 2$. The Φ -Sobolev inequality for $\Phi \leftarrow \Phi_p$ is referred to as the *Beckner inequality* [[Bakry et al., 2014](#), §7.6.2]. Owing to the shared properties between $t \mapsto t^2$, $t \mapsto t \log(t)$ and $\Phi_p(t)$ [[Chafaï, 2004](#), Pg. 5], one might surmise from [Theorems 3.1](#) and [3.2](#) that when ρ satisfies $\Phi\text{SI}(\alpha)$ and $\{P_y\}_{y \in \mathcal{Y}}$ individually satisfy $\Phi\text{SI}(\beta)$ for $\Phi \leftarrow \Phi_p$, then ν and μ would satisfy $\Phi\text{SI}(\zeta)$ and $\Phi\text{SI}(\xi)$ respectively as in [Theorems 3.1](#)

21

Figure 1: Source page 21: the general question and the Beckner specialization.

and [3.2](#) for $\Phi \leftarrow \Phi_p$. However, this has proven challenging to establish. To examine this further, we first note that the duality identity ([Proposition 5.4](#)) holds by virtue of the $\frac{1}{\Phi_p''}$ being concave

$$\mathbb{E}_\pi[(\Phi_p'(f) - \mathbb{E}_\pi[\Phi_p'(f)]) \cdot g] \leq \mathbb{J}_\pi^{\Phi_p}[g] + \mathbb{E}_\pi[g] \cdot (\Phi_p'(\mathbb{E}_\pi[f]) - \mathbb{E}_\pi[\Phi_p'(f)]) + \mathbb{J}_\pi^{\Phi_p^*(\Phi_p')}[f] .$$

The [Var](#) and [MGF](#) conditions arise naturally from this duality identity when $\Phi(t) = t^2$ and $\Phi(t) = t \log(t)$, as one of the second and third terms on the RHS above reduce to 0 and the remaining term non-zero terms are directly related to [Var](#) and [MGF](#) respectively. For Φ_p however, both terms are non-zero. Moreover, we use the Legendre property of the functions t^2 and $t \log(t)$ after instantiating the above identity, which is absent for $\Phi_p(t)$ as $|\Phi_p'(t)| \not\rightarrow \infty$ as $t \rightarrow 0$.

Acknowledgements

Figure 2: Source page 22: the continuation and the obstruction to the source proof route.

If $h \geq 0$ and $\mathbb{E}_P h = 1$, put

$$D_p(h\|P) = \frac{\mathbb{E}_P h^p - 1}{p - 1}.$$

The notation records the reference measure; D_p is the power divergence of the probability law hP from P .

Definition 1 (power-divergence score transport). *For $L \geq 0$, the family (P_y) satisfies $\text{ST}_p(L)$ if, for ρ -almost every y , every probability density h relative to P_y , and every $u \in \mathbb{R}^{d_y}$,*

$$|\mathbb{E}_{P_y}[h \langle u, s_y \rangle]|^2 \leq \frac{2L^2}{p} \|u\|^2 D_p(h\|P_y). \quad (2)$$

This condition is tailored to the homogeneity of Φ_p , not assumed as a black-box miracle: Corollaries 4 and 5 below prove it from two concrete score assumptions used in the source.

3 Proof intuition

The source decomposes the joint Φ -entropy into a conditional entropy and the Φ -entropy of the conditional mean. The only difficult derivative is the score contribution in

$$\nabla_y \mathbb{E}_{P_y} \psi = \mathbb{E}_{P_y} \nabla_y \psi + \mathbb{E}_{P_y} (\psi s_y).$$

Instead of dualizing this covariance, normalize ψ by its conditional mean $M = \mathbb{E}_{P_y} \psi$. Then $h = \psi/M$ is a probability density and the power homogeneity gives the exact identity

$$\text{Ent}_{\Phi_p, P_y}(\psi) = M^p D_p(h\|P_y).$$

Multiplication by $\Phi_p''(M) = pM^{p-2}$ makes the M powers in the squared score covariance match perfectly. Condition (2) therefore bounds the score contribution directly by the conditional Beckner entropy, which the assumed conditional inequality controls by the x -gradient energy. The other derivative term is handled by weighted Cauchy–Schwarz and the concavity of $1/\Phi_p''$. These are precisely the two ingredients needed to reuse the source’s two-scale bookkeeping without any Legendre transform.

4 Full two-scale theorem

For $L \geq 0$ define

$$\zeta(L) = \inf_{C>0} \max\{\beta + \alpha(1 + C^{-1})\beta L^2, \alpha(1 + C)\} \quad (3)$$

$$= \frac{1}{2} \left(\alpha + \beta + \alpha\beta L^2 + \sqrt{4\alpha^2\beta L^2 + (\beta - \alpha + \alpha\beta L^2)^2} \right), \quad (4)$$

$$\xi(L) = \beta + \alpha\beta L^2. \quad (5)$$

Theorem 2 (joint and mixture Beckner inheritance). *Fix $1 < p \leq 2$. Suppose that ρ satisfies $\Phi_p \text{SI}(\alpha)$, every P_y satisfies $\Phi_p \text{SI}(\beta)$ with the same β , and (P_y) satisfies $\text{ST}_p(L)$. Then:*

- (i) *the joint law $\nu = P_y \rho$ satisfies $\Phi_p \text{SI}(\zeta(L))$;*
- (ii) *the mixture $\mu = \int P_y \rho(dy)$ satisfies $\Phi_p \text{SI}(\xi(L))$.*

Proof. We first take a smooth strictly positive admissible function $\psi = \psi(x, y)$ and set

$$M(y) = \mathbb{E}_{P_y} \psi(\cdot, y).$$

The exact entropy decomposition is

$$\text{Ent}_{\Phi_p, \nu}(\psi) = \mathbb{E}_\rho \text{Ent}_{\Phi_p, P_y}(\psi) + \text{Ent}_{\Phi_p, \rho}(M). \quad (6)$$

The first term satisfies

$$\mathbb{E}_\rho \text{Ent}_{\Phi_p, P_y}(\psi) \leq \frac{\beta}{2} \mathbb{E}_\nu [\Phi_p''(\psi) \|\nabla_x \psi\|^2]. \quad (7)$$

Differentiation under the integral gives

$$\nabla M = \mathbb{E}_{P_y} \nabla_y \psi + \mathbb{E}_{P_y} (\psi s_y). \quad (8)$$

Because

$$\frac{1}{\Phi_p''(t)} = \frac{1}{p} t^{2-p}$$

is concave on $(0, \infty)$, weighted Cauchy–Schwarz and Jensen give

$$\begin{aligned} \Phi_p''(M) \|\mathbb{E}_{P_y} \nabla_y \psi\|^2 &\leq \Phi_p''(M) \mathbb{E}_{P_y} [\Phi_p''(\psi) \|\nabla_y \psi\|^2] \mathbb{E}_{P_y} \left[\frac{1}{\Phi_p''(\psi)} \right] \\ &\leq \mathbb{E}_{P_y} [\Phi_p''(\psi) \|\nabla_y \psi\|^2]. \end{aligned} \quad (9)$$

For the score term, put $h = \psi/M$. Then $\mathbb{E}_{P_y} h = 1$ and, exactly,

$$\text{Ent}_{\Phi_p, P_y}(\psi) = M^p D_p(h \| P_y). \quad (10)$$

Taking the supremum over unit u in (2), we obtain

$$\begin{aligned} \Phi_p''(M) \|\mathbb{E}_{P_y} (\psi s_y)\|^2 &= p M^p \|\mathbb{E}_{P_y} (h s_y)\|^2 \\ &\leq 2L^2 M^p D_p(h \| P_y) \\ &= 2L^2 \text{Ent}_{\Phi_p, P_y}(\psi) \\ &\leq \beta L^2 \mathbb{E}_{P_y} [\Phi_p''(\psi) \|\nabla_x \psi\|^2]. \end{aligned} \quad (11)$$

For every $C > 0$, the squared triangle inequality with Young's parameter and (9)–(11) imply

$$\begin{aligned} \Phi_p''(M) \|\nabla M\|^2 &\leq (1 + C) \mathbb{E}_{P_y} [\Phi_p''(\psi) \|\nabla_y \psi\|^2] \\ &\quad + (1 + C^{-1}) \beta L^2 \mathbb{E}_{P_y} [\Phi_p''(\psi) \|\nabla_x \psi\|^2]. \end{aligned} \quad (12)$$

Apply $\Phi_p \text{SI}(\alpha)$ for ρ to M , integrate (12), and combine with (7). This yields

$$\begin{aligned} \text{Ent}_{\Phi_p, \nu}(\psi) &\leq \frac{1}{2} [\beta + \alpha(1 + C^{-1}) \beta L^2] \mathbb{E}_\nu [\Phi_p''(\psi) \|\nabla_x \psi\|^2] \\ &\quad + \frac{\alpha(1 + C)}{2} \mathbb{E}_\nu [\Phi_p''(\psi) \|\nabla_y \psi\|^2]. \end{aligned}$$

Taking the larger coefficient and then the infimum over C proves the joint claim. Equating the two coefficients gives the elementary closed form (4).

For the mixture claim, take $\psi = \psi(x)$ in the entropy decomposition. Now the direct y -derivative term in (8) is zero, so no Young splitting is needed. Equations (7) and (11) give

$$\text{Ent}_{\Phi_p, \mu}(\psi) \leq \frac{\beta + \alpha\beta L^2}{2} \mathbb{E}_\mu[\Phi_p''(\psi) \|\nabla \psi\|^2],$$

which is (5).

The calculation proves the statement for strictly positive test functions. For nonnegative admissible functions, apply it to $\psi + \varepsilon$ and let $\varepsilon \downarrow 0$; the entropy converges and the weighted energy converges by monotone convergence because $p - 2 \leq 0$. Standard truncation and smoothing give the usual finite-energy closure. $\square$

5 Concrete score regimes

The following sharp scalar inequality supplies the exact bounded-score constant.

Lemma 3 (sharp power Pinsker). *For every probability density h relative to P and every $1 < p \leq 2$,*

$$D_p(h\|P) \geq \frac{p}{2} (\mathbb{E}_P|h - 1|)^2. \quad (13)$$

The constant $p/2$ is sharp.

Proof. Since $\mathbb{E}_P h = 1$, write

$$D_p(h\|P) = \mathbb{E}_P f_p(h), \quad f_p(t) = \frac{t^p - pt + p - 1}{p - 1}.$$

Let $A = \{h \geq 1\}$, $\theta = P(A)$, and

$$\delta = \mathbb{E}_P(h - 1)_+ = \mathbb{E}_P(1 - h)_+ = \frac{1}{2} \mathbb{E}_P|h - 1|.$$

The cases $\theta \in \{0, 1\}$ are trivial. Conditional Jensen gives

$$D_p(h\|P) \geq F_\theta(\delta),$$

where

$$F_\theta(t) = \theta f_p\left(1 + \frac{t}{\theta}\right) + (1 - \theta) f_p\left(1 - \frac{t}{1 - \theta}\right).$$

We have $F_\theta(0) = F'_\theta(0) = 0$. Put $a = 1 + t/\theta$, $b = 1 - t/(1 - \theta)$, and $r = 2 - p \in [0, 1]$. On the allowed range of t ,

$$\begin{aligned} F''_\theta(t) &= p \left(\frac{a^{p-2}}{\theta} + \frac{b^{p-2}}{1 - \theta} \right) \\ &= p \left(\frac{1}{\theta a^r} + \frac{1}{(1 - \theta) b^r} \right) \\ &\geq \frac{4p}{\theta a^r + (1 - \theta) b^r} \geq 4p. \end{aligned}$$

The first inequality is the harmonic–arithmetic mean inequality. The second uses concavity of x^r and $\theta a + (1 - \theta) b = 1$. (For $p = 2$, the same conclusion follows directly.) Twice integrating gives $F_\theta(\delta) \geq 2p\delta^2$, which is (13).

For sharpness, on a two-point space with equal masses take $h = 1 + \varepsilon$ and $1 - \varepsilon$. As $\varepsilon \rightarrow 0$, $D_p(h) = \frac{p}{2}\varepsilon^2 + o(\varepsilon^2)$ while $\mathbb{E}|h - 1| = \varepsilon$. $\square$

Corollary 4 (pointwise bounded scores: exact source constants). *If $\|s_y(x)\| \leq B$ for P_y -almost every x and ρ -almost every y , then $\text{ST}_p(B)$ holds. Consequently, the joint and mixture satisfy $\Phi_p \text{SI}(\zeta(B))$ and $\Phi_p \text{SI}(\xi(B))$, respectively, for every $1 < p \leq 2$.*

Proof. For $\|u\| \leq 1$, centering of the score and Lemma 3 give

$$\begin{aligned} |\mathbb{E}_{P_y} h \langle u, s_y \rangle| &= |\mathbb{E}_{P_y} (h - 1) \langle u, s_y \rangle| \\ &\leq B \mathbb{E}_{P_y} |h - 1| \leq B \sqrt{\frac{2}{p} D_p(h \| P_y)}. \end{aligned}$$

Squaring is exactly (2) with $L = B$. □

The next result uses precisely the MGF criterion displayed in the source.

Corollary 5 (source MGF criterion). *Assume that, for every y and u ,*

$$\log \mathbb{E}_{P_y} \exp(\langle u, s_y \rangle) \leq \frac{\bar{L}^2 \|u\|^2}{2}. \quad (14)$$

Then $\text{ST}_p(\sqrt{p} \bar{L})$ holds. Hence the joint and mixture satisfy $\Phi_p \text{SI}(\zeta(\sqrt{p} \bar{L}))$ and $\Phi_p \text{SI}(\xi(\sqrt{p} \bar{L}))$, respectively. Equivalently, replace $\bar{L}^2$ by $p \bar{L}^2$ in the source constant formulas.

Proof. Fix a unit u , write $Z = \langle u, s_y \rangle$, and let $Q = h P_y$. For every $\lambda > 0$, the elementary entropy variational inequality gives

$$\lambda \mathbb{E}_Q Z \leq \text{KL}(Q \| P_y) + \log \mathbb{E}_{P_y} e^{\lambda Z} \leq \text{KL}(Q \| P_y) + \frac{\lambda^2 \bar{L}^2}{2}.$$

Apply the same bound to $-Z$ and optimize in λ to obtain

$$|\mathbb{E}_Q Z|^2 \leq 2 \bar{L}^2 \text{KL}(Q \| P_y). \quad (15)$$

For completeness, Jensen under Q and $\log z \leq z - 1$ give

$$\begin{aligned} (p - 1) \text{KL}(Q \| P_y) &= \mathbb{E}_Q \log h^{p-1} \leq \log \mathbb{E}_Q h^{p-1} \\ &= \log \mathbb{E}_{P_y} h^p \leq \mathbb{E}_{P_y} h^p - 1 = (p - 1) D_p(h \| P_y). \end{aligned}$$

Thus (15) is at most $2 \bar{L}^2 D_p$, which is (2) with $L^2 = p \bar{L}^2$. □

Remark 6 (endpoint behavior). As $p \downarrow 1$, the MGF corollary tends to the source log-Sobolev constant. At $p = 2$ it is conservative by a factor two on the score proxy; the source variance argument gives the sharper Poincaré endpoint. Under pointwise bounded scores, Corollary 4 has the exact source constant for every p , including both endpoints.

6 Why variance alone does not close the intermediate case

The source variance criterion is perfectly adapted to $p = 2$. For $p < 2$ it does not, by itself, imply the score-transport estimate used above.

Proposition 7 (variance-only obstruction). *Fix $1 < p < 2$. There is a centered scalar random variable S with finite variance such that*

$$\sup_{h \geq 0, \mathbb{E} h = 1} \frac{|\mathbb{E}(hS)|^2}{D_p(h \| P)} = \infty.$$

Consequently, no bound on $\text{Var}(S)$ alone can imply (2) with a universal finite constant.

Proof. Choose disjoint events A_n of probabilities $\varepsilon_n > 0$, with $\sum_n \varepsilon_n < 1$, decreasing so fast that $\sum_n n\varepsilon_n^{2-p} < \infty$. Define

$$a_n = \sqrt{n} \varepsilon_n^{-(p-1)/2}, \quad S_0 = \sum_{n \geq 1} a_n \mathbf{1}_{A_n}, \quad S = S_0 - \mathbb{E}S_0.$$

Then

$$\mathbb{E}S_0^2 = \sum_{n \geq 1} n\varepsilon_n^{2-p} < \infty,$$

so S is centered with finite variance. Let $h_n = \mathbf{1}_{A_n}/\varepsilon_n$. Its power divergence is

$$D_p(h_n \| P) = \frac{\varepsilon_n^{1-p} - 1}{p - 1},$$

whereas $\mathbb{E}(h_n S) = a_n - \mathbb{E}S_0$. Since $a_n \rightarrow \infty$,

$$\frac{|\mathbb{E}(h_n S)|^2}{D_p(h_n \| P)} \sim (p - 1)n \rightarrow \infty.$$

Rescaling S fixes its variance at any prescribed positive value without changing the divergence of the ratios. $\square$

Proposition 7 is an obstruction to deriving the theorem from the source’s variance criterion alone. It is *not* asserted to be a counterexample to Beckner inheritance by some different method.

7 Verification, novelty, and limitations

Symbolic checks. The proof was independently re-derived from the entropy decomposition. The identity (10), the factor pM^{p-2} in (11), the two energy coefficients in (3), and the closed form (4) were checked separately. The argument specializes correctly at $p = 2$, and the $p \downarrow 1$ scaling agrees with the entropy endpoint.

Finite checks. The accompanying script tests the sharp power-Pinsker inequality on 441,007 binary instances, the bounded-score estimate on 25,000 random finite laws, and the closed joint constant on 5,000 random parameter triples. It also displays the divergent ratios from Proposition 7. These computations are sanity checks, not proof.

Novelty check. Bounded searches on 11 August 2026 used the exact title, arXiv id, author, Section 6.1 wording, and combinations of “two-scale”, “Beckner”, “mixture”, “conditional score”, and “score transport”. No later paper explicitly answering the question was found. Agrawal–Horel [2] provides relevant general f -divergence/IPM duality but does not state this two-scale theorem. This is evidence of plausible novelty, not a comprehensive priority certification.

Limitations. The factor p in the MGF score proxy is not claimed optimal. The abstract condition (2) is sufficient, not claimed necessary for inheritance. Proposition 7 only rules out a variance-to-score-transport implication. The separate broad question in the source about regularity of Hamiltonian flows is untouched. Finally, the theorem uses the same differentiation and approximation regularity as the source’s Euclidean density setup.

Human-review recommendation. Send to a specialist in functional inequalities. The highest-value checks are (i) every factor of p in (10)–(11), (ii) the lower bound on F''_θ in Lemma 3, and (iii) whether the source’s intended test-function class requires any additional routine approximation statement.

References

- [1] V. Srinivasan, *Understanding two-scale criteria for Poincaré and log-Sobolev inequalities in the Euclidean case through Φ -entropies*, arXiv:2509.15410, 2025.
- [2] R. Agrawal and T. Horel, *Optimal bounds between f -divergences and integral probability metrics*, Journal of Machine Learning Research 22 (2021), no. 128, 1–59; arXiv:2006.05973.